

Aryan Hemant Sahu

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 [Aryan Sahu](#) |  [aryaniscoding](#) | [LeetCode](#) | [Portfolio](#)

EDUCATION

Pune Institute of Computer Technology (PICT)

B.E. in Electronics & Telecommunications (Honours in AI & Machine Learning)

CGPA: 9.68/10

2023 – Present

EXPERIENCE

• iMocha Technologies Pvt. Ltd.

Jan 2026 - present

GenAi Intern

Hybrid

- Architecting production-ready n8n automation workflows to generate, validate, and refine QA content via LLM-based pipelines.
- Engineered internal bulk-processing tools in n8n for automated question proofreading and file schema validation.

• Shoption Pvt. Ltd.

September 2025 - December 2025

IT Intern

Hybrid

- Engineered production-grade multilingual RAG chatbot backend using FastAPI, PostgreSQL+pgvector (HNSW), and Gemini/Cohere, supporting en/hi/mr with 768-dim semantic search with batched translation enhancement.
- Implemented hybrid retrieval pipeline combining dense vector search (cosine ANN), PostgreSQL FTS, and Cohere reranking, processing 200→50→5 passage refinement with real-time CRM enrichment via parallel async HTTP calls.
- Integrated sentiment analysis (TextBlob) and telemetry system tracking per-step latency, cumulative token usage across all LLM operations, CPU/RAM metrics, reducing response latency by 40% through system architecture.

• Aeronica Advance Technologies Pvt. Ltd.

June 2025 - August 2025

Control Systems Intern

Hybrid

- Developed PID, MPC, and LQR algorithms on STM32 for tri-axis stabilization using MPU6050 feedback.
- Optimized control loops to reduce response time by 13%, significantly improving quadcopter stability.

• NxTechWorks Consulting Pvt. Ltd.

May 2025 - June 2025

Data Science Intern

Remote

- Architected a RAG chatbot with Cross-Encoder reranking, improving document retrieval precision by 25% for complex Q&A tasks, and integrated summarization with automatic question-answer generation.
- Produced and deployed n8n workflows that deliver personalized nearby event suggestions via email and automatically sync updates to Google Calendar.

PROJECTS

• AquaMitra: Hybrid RAG + Text-to-SQL

August 2025 - September 2025

Python, FastAPI, Uvicorn, LlamaIndex, HuggingFace, DuckDB, SQLAlchemy + duckdb-engine 

- Built a Hybrid RAG QA assistant with LlamaIndex routing, queries via Text-to-SQL on a single DuckDB table.
- CSV ingested into DuckDB, used case-insensitive text filters and persisted chat logs with executed SQL latency.
- Delivered a FastAPI + Uvicorn backend using SQLAlchemy (duckdb-engine), exposing /api/chat, /api/health, and /api/history and integrating the full RAG/Text-to-SQL pipeline.

• FareFlow: Smart Parking System

Feb 2025 - May 2025

Python, Flask, ESP32-CAM, Plate Recognizer API, SQLite, OpenCV 

- Built an end-to-end parking solution with ESP32-CAM video feed and Flask backend.
- Automated license plate recognition via Plate Recognizer API, achieving 98% accuracy at 0–3m
- Evolved dynamic fare logic calculating fees based on parking duration, eliminating manual billing errors.

SKILLS

- **Programming Languages:** Python, C / C++, MySQL
- **Data Science & ML:** Pandas, Numpy, Matplotlib, Seaborn, Scikitlearn, Tensorflow, NLP, OCR, Transfer learning
- **GenAI & Agentic AI:** N8N, MCP, Langchain
- **Web Technologies:** Flask, Uvicorn, FastAPI
- **Database Systems: Relational:** MySQL, SQL, PostgreSQL, **Vectordb:** Faiss, pgvector
- **DevOps & Version Control:** Docker, Model deployment, Git/Github
- **Other:** Google colab, VS code, Kaggle, Huggingface, Control system algorithms
- **Software development:** OOP, DSA, DBMS, CN, System Design, clean coding

ACHIEVEMENTS

- Demonstrated advanced problem-solving ability by solving over 550 Data Structures and Algorithms problems on LeetCode, Codechef and GfG
- Ranked in the top 45 out of 650+ teams nationwide in the Dream11 gameathon